

Daffodil International University
Faculty of Science & Information Technology
Department of Computer Science & Engineering
Quiz Examination

Course Title: Database Management System	Course Code: CSE 311
Section: 67_P	Marks: 15
Time: 45 Minutes	Date: 20-06-26

Question 1: Scenario-Based ER Diagram [5 Marks]

A coaching center wants to manage its course enrollment system. Each student has a student ID, student name where must include first name and last name, email, and multi valued phone number, age which is derived from birth date. Each course has a course ID, course title, credit, and course fee. A student can enroll in multiple courses, and each course can have multiple students. For every enrollment, the system stores enrollment date and payment status.

Draw a complete ER diagram for the above scenario. Your ER diagram must show the required entities, important attributes, primary keys, relationship, proper cardinality, and how the many-to-many relationship between Student and Course is handled.

Question 2: Database Design Query [3 Marks]

Based on the scenario in Question 1, write SQL queries to create the required database tables. Your design must include suitable table names, proper attributes, primary keys, foreign keys, and suitable data types.

- Create Student table and insert 2 dummy value. [2 mark]
- Create Enrollment table with proper primary key and foreign keys. [1 mark]

Question 3: SQL Queries Without Subquery [7 Marks]

Use the following tables to answer the SQL questions.

Student Table

student_id	student_name	email	phone
101	Rahim Ahmed	rahim@gmail.com	01711111111
102	Karim Hasan	karim@gmail.com	01722222222
103	Nusrat Jahan	nusrat@gmail.com	01733333333
104	Tanvir Islam	tanvir@gmail.com	01744444444
105	Rafi Khan	rafi@gmail.com	01755555555

Course Table

course_id	course_title	credit	fee
1	Database Management System	3	5000
2	Artificial Intelligence	3	6000
3	Web Development	2	4000
4	Computer Network	3	5500

Enrollment Table

enrollment_id	student_id	course_id	enrollment_date	payment_status
1	101	1	2026-06-01	Paid
2	102	1	2026-06-02	Unpaid
3	103	2	2026-06-03	Paid
4	104	3	2026-06-04	Paid
5	101	2	2026-06-05	Unpaid
6	105	4	2026-06-06	Paid

Write SQL queries for the following:

- Show all students whose names start with R. [1 mark]
- Show all courses where the fee is between 4000 and 5500. [1 mark]
- Show all enrollments where payment status is either Paid or Unpaid using IN. [1 mark]
- Show all courses with credit equal to 3. [1 mark]
- Count how many students enrolled in each course using GROUP BY. [1.5 marks]
- Show only those courses where total enrollment is greater than 1 using GROUP BY and HAVING. [1.5 marks]